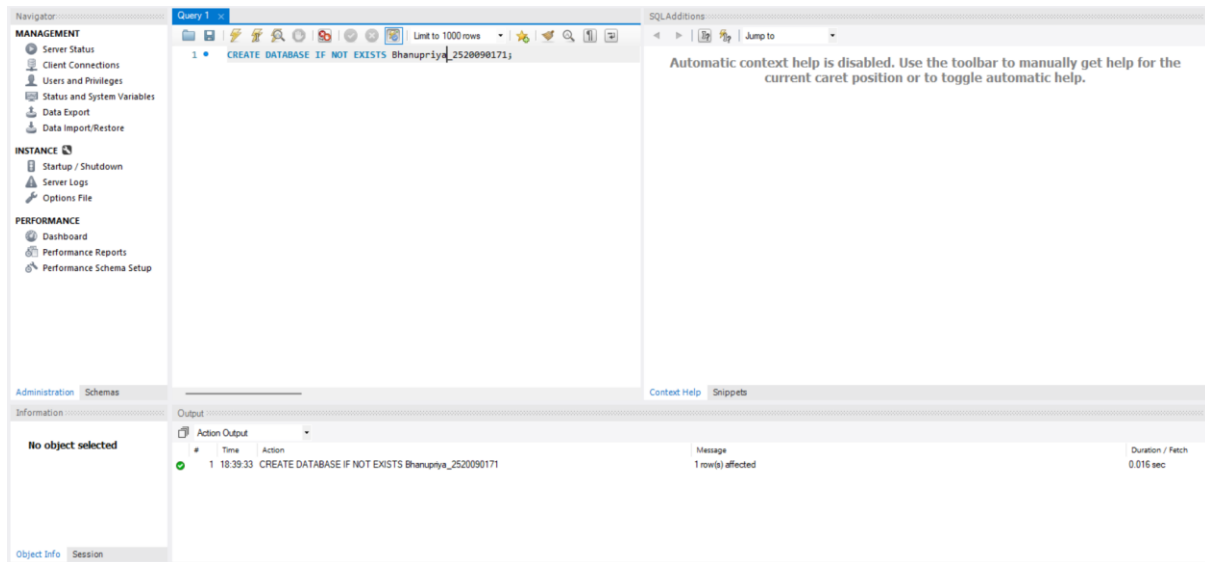


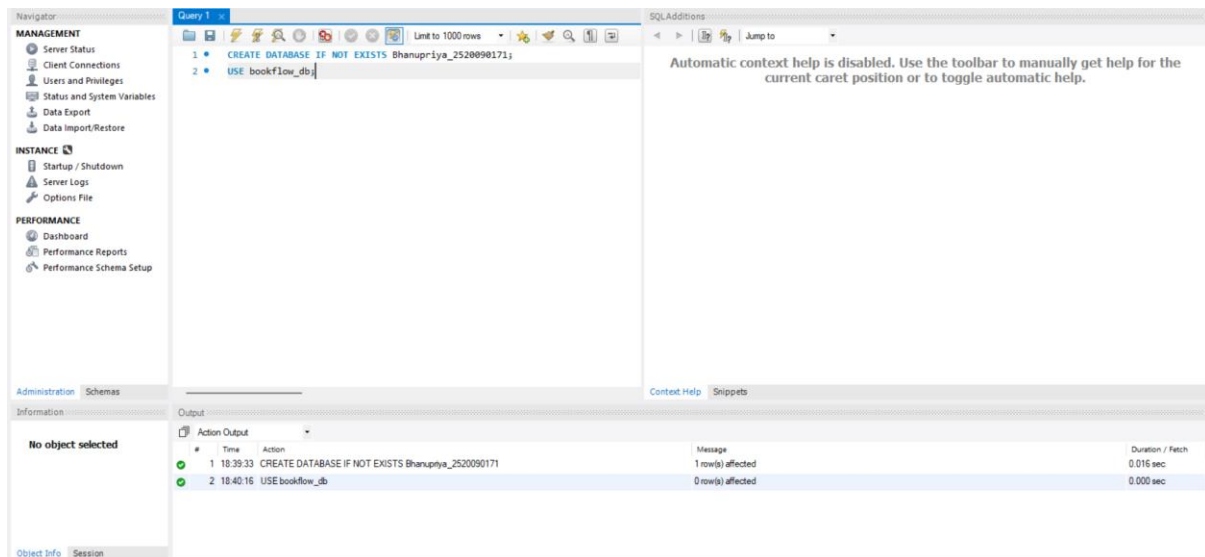
WEEK-1

Step 1 Database Setup

Query 1: Create the database

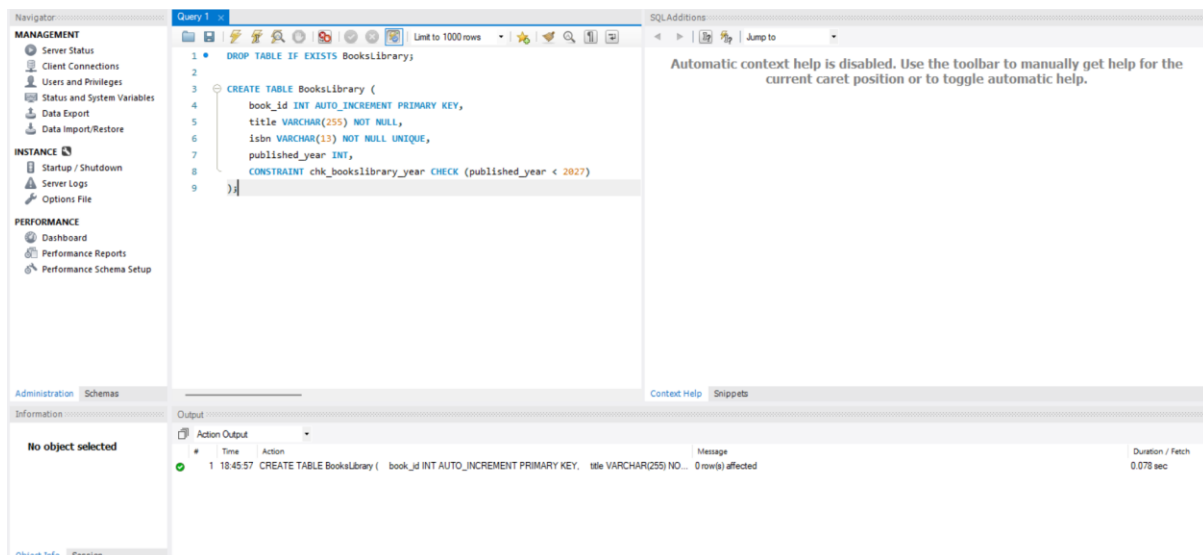


Query 2: Select the database

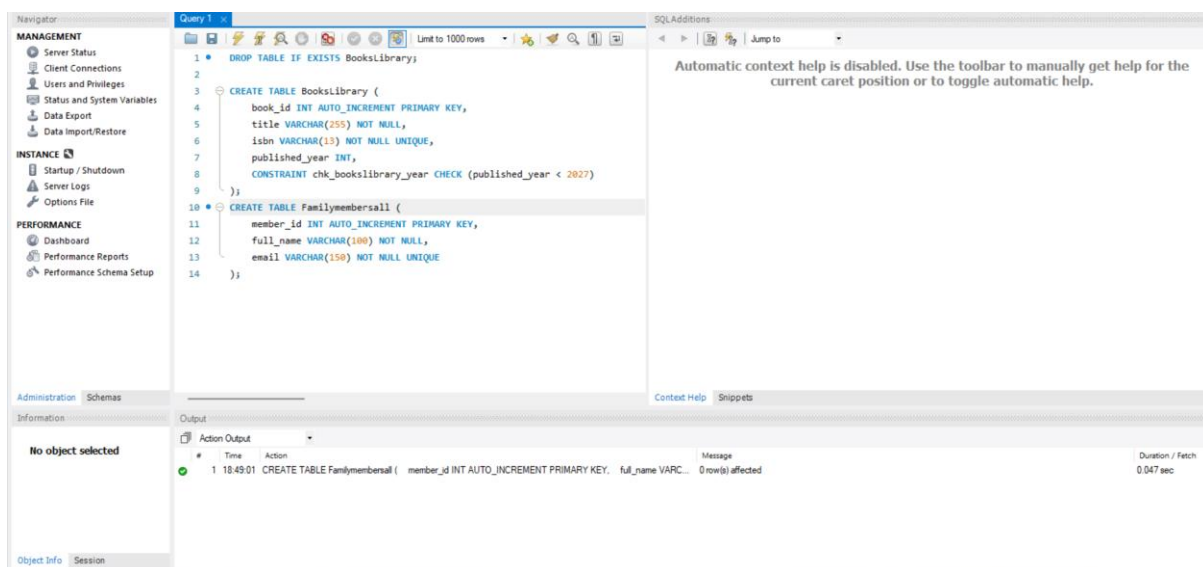


Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table



Query 4: Create the members table



Query 5 (verification): Inspect the table structures

Query 1

```

10 CREATE TABLE Familymembersall (
11     member_id INT AUTO_INCREMENT PRIMARY KEY,
12     full_name VARCHAR(100) NOT NULL,
13     email VARCHAR(150) NOT NULL UNIQUE
14 );
15 DESCRIBE books;

```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI		auto_increment
title	varchar(255)	NO			
isbn	varchar(13)	NO	UNI		
published_year	int	YES			

Result Grid

Output

#	Time	Action	Message	Duration / Fetch
1	18:49:01	CREATE TABLE Familymembersall (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.047 sec
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec

Step 3 Initial Seeding (Valid Data)

Query 6: Insert 3 books

Query 1

```

1 DROP TABLE IF EXISTS BooksLibrary;
2
3 CREATE TABLE BooksLibrary (
4     book_id INT AUTO_INCREMENT PRIMARY KEY,
5     title VARCHAR(255) NOT NULL,
6     isbn VARCHAR(13) NOT NULL UNIQUE,
7     published_year INT,
8     CONSTRAINT chk_bookslibrary_year CHECK (published_year < 2027)
9 );
10 CREATE TABLE Familymembersall (
11     member_id INT AUTO_INCREMENT PRIMARY KEY,
12     full_name VARCHAR(100) NOT NULL,
13     email VARCHAR(150) NOT NULL UNIQUE
14 );
15 DESCRIBE books;
16 INSERT INTO books (title, isbn, published_year) VALUES
17 ('The Alchemist', '9780061122415', 1988),
18 ('Clean Code', '9780132350884', 2008),
19 ('Atomic Habits', '9780735211292', 2018);

```

Output

#	Time	Action	Message	Duration / Fetch
1	18:49:01	CREATE TABLE Familymembersall (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.047 sec
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
3	18:50:41	INSERT INTO books title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Query 7: Insert 3 members

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

4  book_id INT AUTO_INCREMENT PRIMARY KEY,
5  title VARCHAR(255) NOT NULL,
6  isbn VARCHAR(13) NOT NULL UNIQUE,
7  published_year INT,
8  CONSTRAINT chk_booklibrary_year CHECK (published_year < 2027)
9  );
10 CREATE TABLE Familymembersall (
11     member_id INT AUTO_INCREMENT PRIMARY KEY,
12     full_name VARCHAR(100) NOT NULL,
13     email VARCHAR(150) NOT NULL UNIQUE
14 );
15 DESCRIBE books;
16 INSERT INTO books (title, isbn, published_year) VALUES
17 ('The Alchemist', '9780061122415', 1988),
18 ('Clean Code', '9780132358884', 2008),
19 ('Atomic Habits', '9780735211292', 2018);
20 INSERT INTO members (full_name, email) VALUES
21 ('Anil Kumar', 'anil.kumar@example.com'),
22 ('Priya Sharma', 'priya.sharma@example.com'),
23 ('Ravi Verma', 'ravi.verma@example.com');

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:49:01	CREATE TABLE Familymembersall (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.047 sec
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
3	18:50:41	INSERT INTO books title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
4	18:51:17	INSERT INTO members full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec

Navigator

MANAGEMENT

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PERFORMANCE

- Dashboard
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- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

18 ('Clean Code', '9780132358884', 2008),
19 ('Atomic Habits', '9780735211292', 2018);
20 INSERT INTO members (full_name, email) VALUES
21 ('Anil Kumar', 'anil.kumar@example.com'),
22 ('Priya Sharma', 'priya.sharma@example.com'),
23 ('Ravi Verma', 'ravi.verma@example.com');
24 SELECT * FROM books;

```

Result Grid

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	Clean Code	9780132358884	2008
3	Atomic Habits	9780735211292	2018

books 2 x

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:49:01	CREATE TABLE Familymembersall (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.047 sec
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
3	18:50:41	INSERT INTO books title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
4	18:51:17	INSERT INTO members full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	18:51:55	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 4 Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

Navigator

MANAGEMENT

- Server Status
- Client Connections
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- Status and System Variables
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INSTANCE

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- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

Query 1

```

7  published_year INT,
8  CONSTRAINT chk_booklibrary_year CHECK (published_year < 2027)
9  );
10 CREATE TABLE Familymembersall (
11     member_id INT AUTO_INCREMENT PRIMARY KEY,
12     full_name VARCHAR(100) NOT NULL,
13     email VARCHAR(150) NOT NULL UNIQUE
14 );
15 DESCRIBE books;
16 INSERT INTO books (title, isbn, published_year) VALUES
17 ('The Alchemist', '9780061122415', 1988),
18 ('Clean Code', '9780132358884', 2008),
19 ('Atomic Habits', '9780735211292', 2018);
20 INSERT INTO members (full_name, email) VALUES
21 ('Anil Kumar', 'anil.kumar@example.com'),
22 ('Priya Sharma', 'priya.sharma@example.com'),
23 ('Ravi Verma', 'ravi.verma@example.com');
24 SELECT * FROM books;
25 INSERT INTO books (title, isbn, published_year)
26 VALUES ('Fake Copy', '9780061122415', 2000);

```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

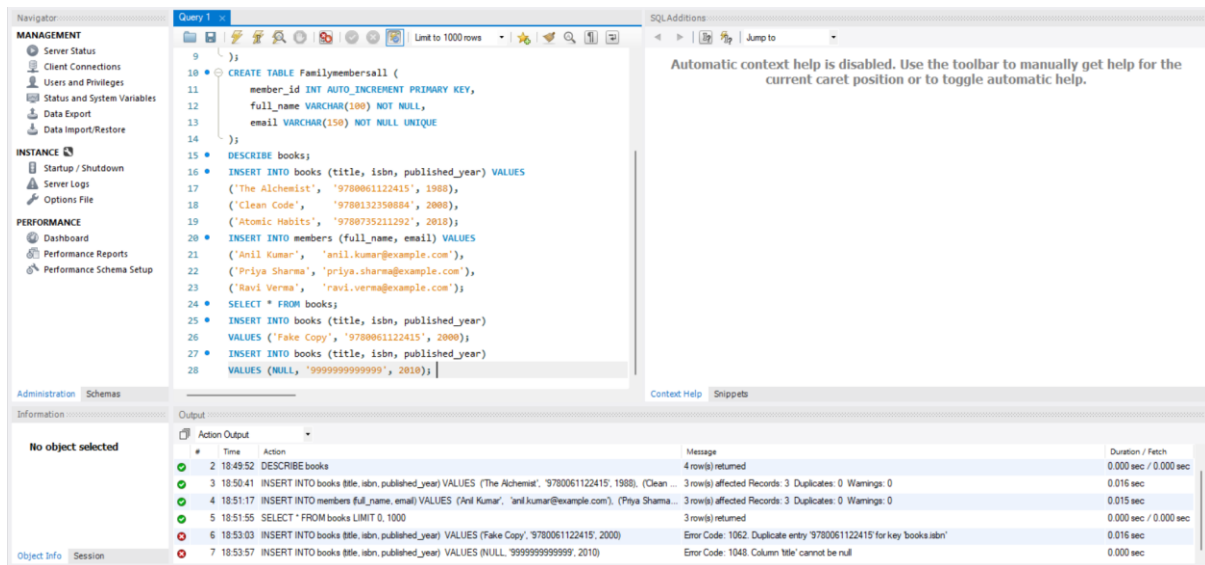
Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:49:01	CREATE TABLE Familymembersall (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARC...	0 row(s) affected	0.047 sec
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
3	18:50:41	INSERT INTO books title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean C...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
4	18:51:17	INSERT INTO members full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	18:51:55	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	18:53:03	INSERT INTO books title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books isbn'	0.016 sec

Test 2: NULL title (NOT NULL violation)



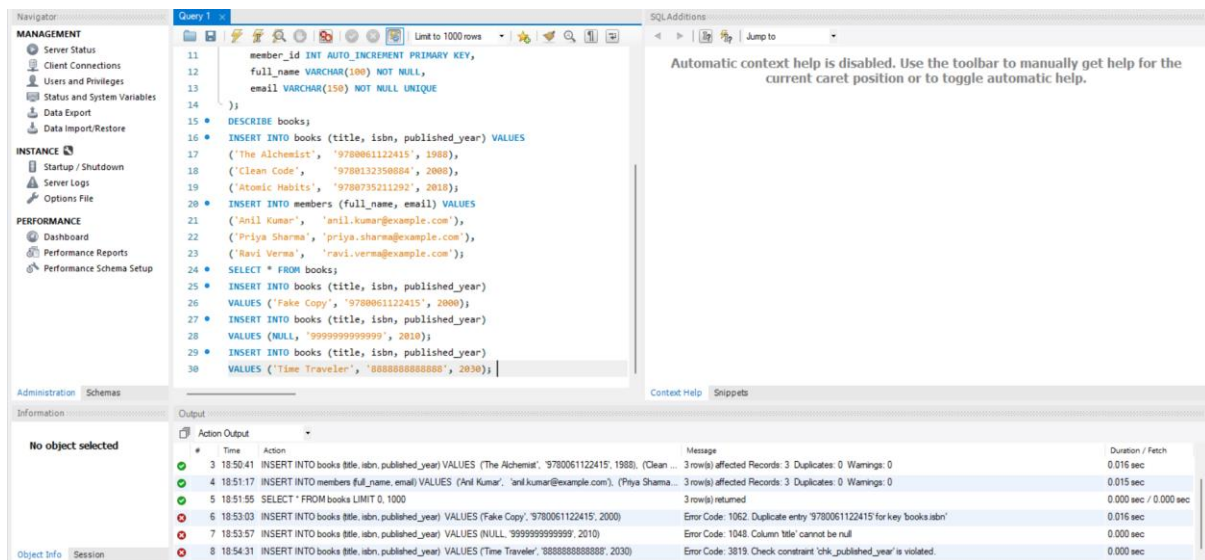
The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Navigation' tree with 'Management', 'Instance', and 'Performance' sections. The 'Query 1' window shows the following SQL script:

```
9
10
11 CREATE TABLE Familymembersall (
12     member_id INT AUTO_INCREMENT PRIMARY KEY,
13     full_name VARCHAR(100) NOT NULL,
14     email VARCHAR(150) NOT NULL UNIQUE
15 );
16
17 DESCRIBE books;
18
19 INSERT INTO books (title, isbn, published_year) VALUES
20 ('The Alchemist', '9780061122415', 1988),
21 ('Clean Code', '9780132350884', 2008),
22 ('Atomic Habits', '9780735211292', 2018);
23
24 INSERT INTO members (full_name, email) VALUES
25 ('Anil Kumar', 'anil.kumar@example.com'),
26 ('Priya Sharma', 'priya.sharma@example.com'),
27 ('Ravi Verma', 'ravi.verma@example.com');
28
29 SELECT * FROM books;
30
31 INSERT INTO books (title, isbn, published_year)
32 VALUES ('Fake Copy', '9780061122415', 2000);
33
34 INSERT INTO books (title, isbn, published_year)
35 VALUES (NULL, '999999999999', 2010);
```

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
2	18:49:52	DESCRIBE books	4 row(s) returned	0.000 sec / 0.000 sec
3	18:50:41	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Atomic Habits', '9780735211292', 2018);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
4	18:51:17	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', 'priya.sharma@example.com'), ('Ravi Verma', 'ravi.verma@example.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	18:51:55	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	18:53:03	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.016 sec
7	18:53:57	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '999999999999', 2010);	Error Code: 1048. Column 'title' cannot be null	0.000 sec

Test 3: Future publication year (CHECK violation)



The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Navigation' tree with 'Management', 'Instance', and 'Performance' sections. The 'Query 1' window shows the following SQL script:

```
11
12 member_id INT AUTO_INCREMENT PRIMARY KEY,
13 full_name VARCHAR(100) NOT NULL,
14 email VARCHAR(150) NOT NULL UNIQUE
15 );
16
17 DESCRIBE books;
18
19 INSERT INTO books (title, isbn, published_year) VALUES
20 ('The Alchemist', '9780061122415', 1988),
21 ('Clean Code', '9780132350884', 2008),
22 ('Atomic Habits', '9780735211292', 2018);
23
24 INSERT INTO members (full_name, email) VALUES
25 ('Anil Kumar', 'anil.kumar@example.com'),
26 ('Priya Sharma', 'priya.sharma@example.com'),
27 ('Ravi Verma', 'ravi.verma@example.com');
28
29 SELECT * FROM books;
30
31 INSERT INTO books (title, isbn, published_year)
32 VALUES ('Fake Copy', '9780061122415', 2000);
33
34 INSERT INTO books (title, isbn, published_year)
35 VALUES (NULL, '999999999999', 2010);
36
37 INSERT INTO books (title, isbn, published_year)
38 VALUES ('Time Traveler', '888888888888', 2030);
```

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
3	18:50:41	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Atomic Habits', '9780735211292', 2018);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
4	18:51:17	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma', 'priya.sharma@example.com'), ('Ravi Verma', 'ravi.verma@example.com');	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	18:51:55	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	18:53:03	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.016 sec
7	18:53:57	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '999999999999', 2010);	Error Code: 1048. Column 'title' cannot be null	0.000 sec
8	18:54:31	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '888888888888', 2030);	Error Code: 3819. Check constraint 'chk_published_year' is violated.	0.000 sec

Test 4: Duplicate email (UNIQUE violation)

MANAGEMENT

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INSTANCE

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PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Query 1

```
13 email VARCHAR(150) NOT NULL UNIQUE
14 };
15
16 • DESCRIBE books;
17 • INSERT INTO books (title, isbn, published_year) VALUES
18 ('The Alchemist', '9780061122415', 1988),
19 ('Clean Code', '9780132359884', 2008),
20 ('Atomic Habits', '9780735211292', 2018);
21 • INSERT INTO members (full_name, email) VALUES
22 ('Anil Kumar', 'anil.kumar@example.com'),
23 ('Priya Sharma', 'priya.sharma@example.com'),
24 ('Ravi Verma', 'ravi.verma@example.com');
25 • SELECT * FROM books;
26 • INSERT INTO books (title, isbn, published_year)
27 VALUES ('Fake Copy', '9780061122415', 2000);
28 • INSERT INTO books (title, isbn, published_year)
29 VALUES (NULL, '999999999999', 2010);
30 • INSERT INTO books (title, isbn, published_year)
31 VALUES ('Time Traveler', '888888888888', 2030);
32 • INSERT INTO members (full_name, email)
33 VALUES ('Anil Clone', 'anil.kumar@example.com');
```

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	18:51:17	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com') (Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.015 sec
5	18:51:55	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
6	18:53:03	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'	0.016 sec
7	18:53:57	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null	0.000 sec
8	18:54:31	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.	0.000 sec
9	18:55:33	INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'	0.031 sec